

Title

Comparative Efficacy of Single-Modality Versus Multimodal Neuroimaging for Brain-Machine Interface Rehabilitation Systems: Protocol for a Controlled Training Study

Abstract

Brain-computer interfaces coupled with muscle stimulation represent promising adjuncts to post-stroke recovery protocols. However, uncertainty remains regarding optimal signal acquisition strategies. This protocol describes a blinded comparative trial investigating whether combined electroencephalographic and optical neuroimaging provides advantages over electroencephalography alone in supporting motor control via electrical muscle stimulation. Objectives: To compare classification performance, user experience, and physiological measures across single-modality and dual-modality signal processing pipelines. Methods: Healthy volunteer participants will undergo identical training regimens with either combined or single-modality signal interfaces over a three-day period. Outcome measurement focuses on real-time classification accuracy, subjective agency ratings, attentional load assessment, and comfort metrics. We hypothesize that multimodal acquisition will yield superior performance; secondary objectives include channel optimization and feasibility assessment.